STAT332: Assignment 1 Question 5

Ashley Chui (21003047)

5. An instructor at the University of Florida conducted a small survey of the students in his introductory to statistics class. To this end, he selected a simple random sample of 57 students from the 250 students in his class. The dataset file contains 11 variables, but this question is concerned with the 3 variables given in the table below.

```
library("survey")

data <- read.table("classurv.txt", header = TRUE)

# remove missing GPA values coded as -9
data$GPA[data$GPA == -9] <- NA

# Define the survey design
dsgn <- svydesign(ids=~1, probs=c(57/350), fpc=rep(350, nrow(data)), data=data)
df0 <- degf(dsgn)
```

(a) Estimate the average GPA of all students

```
a <- svymean(~GPA, dsgn, na.rm=TRUE)
confint(a, df = df0)
```

```
##        2.5 %   97.5 %
## GPA 2.905407 3.106676
```

```
a
```

```
##      mean     SE
## GPA 3.006 0.0502
```

The point estimate for the average GPA of all students is 1.11 with a 95% confidence interval $[0.0347, 2.1860]$.

(b)-(c) Estimate average GPA of male/female students

```
b_c <- svyby(~GPA, ~gender, dsgn, svymean, na.rm = TRUE)
confint(b_c, df = df0)
```

```
##      2.5 %   97.5 %
## 1 2.922683 3.175578
## 2 2.813588 3.119212
```

```r
b_c
```

```
##   gender     GPA         se
## 1      1 3.04913 0.06312168
## 2      2 2.96640 0.07628240
```

The point estimate for the average GPA of male students is 3.049 with a 95% confidence interval $[2.9227, 3.1756]$.\ The point estimate for the average GPA of female students is 2.966 with a 95% confidence interval $[2.8316, 3.1192]$.\

(d) Estimate the proportions of male/female students

```r
d <- svymean(~factor(gender), dsgn)
confint(d, df = df0)
```

```
##                     2.5 %    97.5 %
## factor(gender)1 0.2833467 0.5236709
## factor(gender)2 0.4763291 0.7166533
```

```r
d
```

```
##                    mean   SE
## factor(gender)1 0.40351 0.06
## factor(gender)2 0.59649 0.06
```

The estimated proportion of male and female students is 40.35% and 59.65% respectively. The 95% confidence interval for the male proportion is $[28.33%, 52.37%], while for the female proportion it is $[47.63\%, 71.67\%]$.

(e) Estimate the proportions of students that are freshmen/sophomore/junior/senior/other

```r
e_1 <- svyciprop(~I(year == 1), dsgn, method = "logit") # freshmen
e_2 <- svyciprop(~I(year == 2), dsgn, method = "logit") # sophomore
e_3 <- svyciprop(~I(year == 3), dsgn, method = "logit") # junior
e_4 <- svyciprop(~I(year == 4), dsgn, method = "logit") # senior
e_5 <- svyciprop(~I(year == 5), dsgn, method = "logit") # other
```

(f) Amongst freshmen, what are the proportions of male/female students

```r
freshmen <- subset(dsgn, year == 1)
f_male <- svyciprop(~I(gender == 1), freshmen, method = "logit")
f_female <- svyciprop(~I(gender == 2), freshmen, method = "logit")
f_male
```

```
##                        2.5%  97.5%
## I(gender == 1) 0.1250 0.0137 0.5957
```

```r
f_female
```

```
##                      2.5% 97.5%
## I(gender == 2) 0.875 0.404 0.986
```

(g) Repeat (f), but for seniors

```r
seniors <- subset(dsgn, year == 4)
g_male <- svyciprop(~I(gender == 1), seniors, method = "logit")
g_female <- svyciprop(~I(gender == 2), seniors, method = "logit")
g_male
```

```
##                       2.5% 97.5%
## I(gender == 1) 0.600 0.346 0.810
```

```r
g_female
```

```
##                       2.5% 97.5%
## I(gender == 2) 0.400 0.190 0.654
```

(h) Amongst the top performers (i.e. GPA >= 3.2), what are the proportions of male/female students

```r
top_perf <- subset(dsgn, GPA >= 3.2)
h_male <- svyciprop(~I(gender == 1), top_perf, method = "logit")
h_female <- svyciprop(~I(gender == 2), top_perf, method = "logit")
h_male
```

```
##                       2.5% 97.5%
## I(gender == 1) 0.500 0.285 0.715
```

```r
h_female
```

```
##                       2.5% 97.5%
## I(gender == 2) 0.500 0.285 0.715
```